

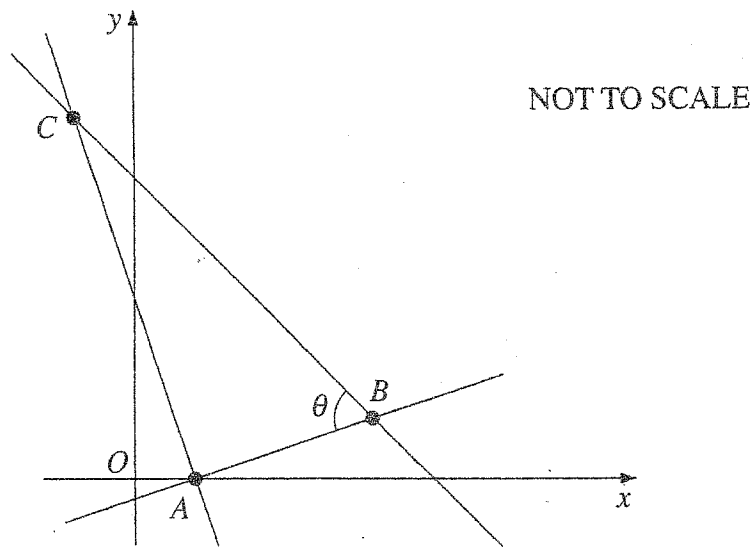
Time : 70 minutes

Instructions :

- * Complete the test on your own paper.
- * Show all necessary working.
- * Marks will be deducted for careless or badly arranged work.

QUESTION 1. (12 marks)

The diagram shows points A(1,0), B(4,1) and C(-1,6) in the Cartesian plane. Angle ABC is θ .



Copy or trace this diagram.

- | | |
|--|---|
| (a) Show that A and C lie on the line $3x + y = 3$. | 2 |
| (b) Show that the gradient of AB is $\frac{1}{3}$. | 1 |
| (c) Show that the length of AB is $\sqrt{10}$ units. | 1 |
| (d) Show that AB and AC are perpendicular. | 1 |
| (e) Find $\tan \theta$. | 2 |
| (f) Find the equation of the circle with centre A that passes through B. | 2 |
| (g) The point D is not shown on the diagram. The point D lies on the line $3x + y = 3$ between A and C, and $AD = AB$. Find the coordinates of D. | 2 |
| (h) On your diagram, shade the region satisfying the inequality $3x + y \leq 3$. | 1 |

QUESTION 2 (7 marks)

- (a) Find the equation of the line through the point of intersection of $2y - 3 = 5x$ and $x + y + 1 = 0$ which is perpendicular to $4x + 2y - 3 = 0$. 4
- (b) Find the perpendicular distance from the point $(1, -2)$ to the line $3x - 2y - 6 = 0$. 3

QUESTION 3 (13 marks)

- (a) Find : (i) $\lim_{x \rightarrow 0} \frac{x^2 + 3x}{x}$ (ii) $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 5x + 6}$ (iii) $\lim_{x \rightarrow \infty} \frac{7x^3 - 6x^2 + 3}{3x^3 + 2x^2 - 4}$ 8
- (b) For $f(x) = x^2 + 3$ find $f(x+h) - f(x)$ 2
- (c) Differentiate $x^2 + 3$ from first principles. 3

QUESTION 4 (20 marks)

- (a) Differentiate : (i) $4x^3 - 2x^2 + 3x - 6$ (ii) $(2x+3)(3x^2-1)$ (iii) $\frac{5}{\sqrt{4+x}}$ 15
- (iv) $(4x+3)^7$ (v) $\frac{2x+1}{x-3}$ (vi) $\frac{2x^5+7x-1}{2x^2}$
- (b) If $f(t) = 5t - 10t^2$, find : 5
- (i) $f(2)$ (ii) $f'(2)$

QUESTION 5 (12 marks)

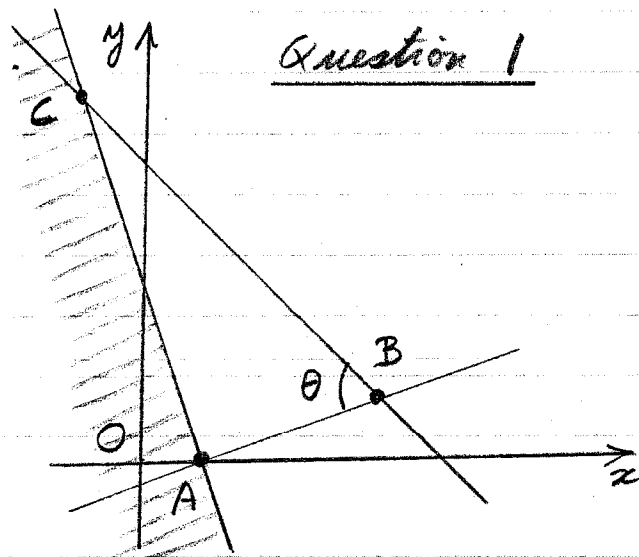
- (a) Find the equation of the normal to the curve $y = x^3 + 2x^2 - 4x - 1$ at the point $(-1, 7)$. 4
- (b) Find any values of p if the tangent to the curve $y = px^2 - 6x - 3$ is parallel to the line $5x - 3y + 2 = 0$ at the point where $x = -3$. 4
- (c) Show that the following three lines are concurrent. 4

$$x - y = 2$$

$$2x + y = 1$$

$$3x + 2y = 1$$

Solutions to Yr. 11 (2U) Task 4 Aug. 2003



(a) $A(1,0)$; $C(-1,6)$, $3x+y=3$

For $A(1,0)$

LHS = $3 \times 1 + 0 = 3 = \text{RHS}$

For $C(-1,6)$

LHS = $3 \times -1 + 6 = 3 = \text{RHS}$

$\therefore A \text{ \& } C \text{ lie on } 3x+y=3$

(b) $m_{AB} = \frac{1-0}{4-1} = \frac{1}{3}$ ①

(c) $AB = \sqrt{(1-0)^2 + (4-1)^2} = \sqrt{10} \text{ units}$ ①

(d) $m_{AB} = \frac{1}{3}$, $m_{AC} = \frac{6-0}{-1-1} = -\frac{6}{2} = -3$

$\therefore m_{AB} \times m_{AC} = \frac{1}{3} \times -3 = -1$

$\therefore AB \perp AC$ ①

(e) $\tan \theta = \frac{AC}{AB}$

$AC = \sqrt{(-1-1)^2 + (6-0)^2} = \sqrt{40}$
 $= 2\sqrt{10}$

$\therefore \tan \theta = \frac{2\sqrt{10}}{\sqrt{10}} = 2$ ②

(f) $A(1,0)$; $AB = \sqrt{10}$

$\therefore (x-1)^2 + (y-0)^2 = (\sqrt{10})^2$

$\therefore (x-1)^2 + y^2 = 10$ ②

(g) D is mid-point of AC

$D\left(\frac{1-1}{2}, \frac{0+6}{2}\right)$

i.e. $D(0,3)$ ②

(h) see diagram. ①

Question 2

(a) $(5x-2y+3) + k(x+y+1) = 0$

$5x+kx-2y+ky+3+k=0$

$x(k+5) + y(k-2) + (k+3) = 0$

$y(k-2) = -x(k+5) - (k+3)$

$\therefore m = \frac{-(k+5)}{k-2} = \frac{k+5}{2-k}$

$4x+2y-3=0$

$\therefore 2y = -4x+3$

$\therefore y = \frac{-4x+3}{2}$

$= -2x + \frac{3}{2}$

$\therefore m = -2$

$\therefore \frac{k+5}{2-k} \times -2 = -1$

$2(k+5) = 2-k$

$2k+10 = 2-k$

$2k+k = 2-10$

$3k = -8$

$\therefore k = -\frac{8}{3}$

$$\begin{aligned} \therefore \text{eqn. of line is} \\ 5x - 2y + 3 - \frac{8}{3}(x + y + 1) &= 0 \\ 15x - 6y + 9 - 8x - 8y - 8 &= 0 \\ \therefore 7x - 14y + 1 &= 0. \quad (4) \end{aligned}$$

$$\begin{aligned} (b) \quad d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ &= \frac{|3(1) - 2(2) - 6|}{\sqrt{3^2 + (-2)^2}} \\ &= \frac{|3 + 4 - 6|}{\sqrt{13}} \\ &= \frac{1}{\sqrt{13}} \text{ units} \quad (3) \end{aligned}$$

Question 3

$$\begin{aligned} (a) \quad (i) \quad \lim_{x \rightarrow 0} \frac{x^2 + 3x}{x} \\ &= \lim_{x \rightarrow 0} \frac{x(x+3)}{x} \quad (2) \\ &= \lim_{x \rightarrow 0} (x+3) = 3. \end{aligned}$$

$$\begin{aligned} (ii) \quad \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 5x + 6} \\ &= \lim_{x \rightarrow 2} \frac{(x+1)(x-2)}{(x-3)(x-2)} \\ &= \lim_{x \rightarrow 2} \frac{x+1}{x-3} \\ &= -3. \quad (3) \end{aligned}$$

$$\begin{aligned} (iii) \quad \lim_{x \rightarrow \infty} \frac{7x^3 - 6x^2 + 3}{3x^3 + 2x^2 - 4} \\ &= \lim_{x \rightarrow \infty} \frac{7 - \frac{6}{x} + \frac{3}{x^3}}{3 + \frac{2}{x} - \frac{4}{x^3}} \end{aligned}$$

$$\begin{aligned} &= \frac{7}{3} \text{ or } 2\frac{1}{3} \\ &\text{since } \lim_{x \rightarrow \infty} \frac{k}{x} = 0 \\ &\text{and } \lim_{x \rightarrow \infty} \frac{k}{x^3} = 0. \quad (3) \\ & (k \text{ a constant}). \end{aligned}$$

$$\begin{aligned} (b) \quad f(x) &= x^2 + 3 \\ \therefore f(x+h) &= (x+h)^2 + 3 \\ &= x^2 + 2hx + h^2 + 3 \\ \therefore f(x+h) - f(x) &= 2hx + h^2 \quad (2) \end{aligned}$$

$$\begin{aligned} (c) \quad \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2hx + h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h} \\ &= 2x. \quad (3) \end{aligned}$$

Question 4

$$\begin{aligned} (a) \quad (i) \quad y &= 4x^3 - 2x^2 + 3x - 6 \\ \therefore \frac{dy}{dx} &= 12x^2 - 4x + 3. \quad (2) \end{aligned}$$

$$\begin{aligned} (ii) \quad y &= (2x+3)(3x^2-1) \\ &= 6x^3 - 2x + 9x^2 - 3 \\ \therefore \frac{dy}{dx} &= 18x^2 + 18x - 2 \quad (2) \end{aligned}$$

Question 4 (ctd.)

$$(iii) y = \frac{5}{\sqrt{4+x}} = 5(4+x)^{-\frac{1}{2}}$$

$$\therefore \frac{dy}{dx} = 5 \cdot -\frac{1}{2}(4+x)^{-\frac{3}{2}}$$

$$= \frac{-5}{2(4+x)^{\frac{3}{2}}}$$

$$= \frac{-5}{2\sqrt{(4+x)^3}} \quad (3)$$

$$(iv) y = (4x+3)^7$$

$$\therefore \frac{dy}{dx} = 7(4x+3)^6 \cdot 4$$

$$= 28(4x+3)^6 \quad (2)$$

$$(v) y = \frac{2x+1}{x-3}$$

$$\therefore \frac{dy}{dx} = \frac{(x-3)2 - (2x+1)1}{(x-3)^2}$$

$$= \frac{2x-6-2x-1}{(x-3)^2}$$

$$= \frac{-7}{(x-3)^2} \quad (3)$$

$$(vi) y = \frac{2x^5 + 7x - 1}{2x^2}$$

$$= x^3 + \frac{7}{2}x^{-1} - \frac{1}{2}x^{-2}$$

$$\therefore \frac{dy}{dx} = 3x^2 - \frac{7}{2}x^{-2} + x^{-3}$$

$$= 3x^2 - \frac{7}{2x^2} + \frac{1}{x^3} \quad (3)$$

$$(b) f(t) = 5t - 10t^2$$

$$(i) f(2) = 5(2) - 10(2^2)$$

$$= 10 - 40$$

$$= -30. \quad (2)$$

$$(ii) f'(t) = 5 - 20t$$

$$\therefore f'(2) = 5 - 40$$

$$= -35. \quad (3)$$

Question 5

$$(a) y = x^3 + 2x^2 - 4x - 1$$

$$\frac{dy}{dx} = 3x^2 + 4x - 4$$

$$\text{at } (-1, 7) \quad \frac{dy}{dx} = 3(-1)^2 + 4(-1) - 4$$

$$= -5$$

$\therefore$ gradient of normal is $\frac{1}{5}$

$\therefore$ eqn. of normal is

$$y - 7 = \frac{1}{5}(x + 1)$$

$$5y - 35 = x + 1$$

$$\therefore x - 5y + 36 = 0. \quad (4)$$

$$(b) y = px^2 - 6x - 3$$

$$\therefore \frac{dy}{dx} = 2px - 6$$

$$\text{at } x = -3, \quad \frac{dy}{dx} = -6p - 6$$

$$5x - 3y + 2 = 0$$

$$\therefore 3y = 5x + 2$$

$$\therefore y = \frac{5}{3}x + \frac{2}{3}$$

$$\therefore m = \frac{5}{3}$$

$$\therefore \frac{5}{3} = -6p - 6$$

$$5 = -18p - 18$$

$$18p = -23$$

$$\therefore p = -\frac{23}{18} \text{ or } -1\frac{5}{18}$$

(4)

$$(c) \quad x - y = 2 \quad \dots (1)$$

$$2x + y = 1 \quad \dots (2)$$

$$3x + 2y = 1 \quad \dots (3)$$

$$(1) + (2) \Rightarrow 3x = 3$$

$$\therefore x = 1$$

$$\text{From (1), } 1 - y = 2$$

$$\therefore y = -1$$

Substitute $x=1, y=-1$ into (3)

$$\therefore \text{LHS} = 3(1) + 2(-1)$$

$$= 3 - 2$$

$$= 1$$

$$= \text{RHS}$$

(4)

$\therefore$ 3 lines above go through

$$x=1, y=-1$$

ie 3 lines are concurrent.